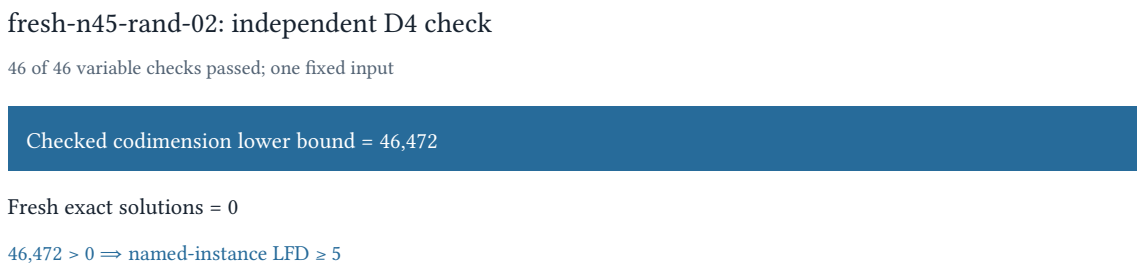
Certified n45 random02: LFD at least five

The independent D4 checker accepted all 46 variable multiplication checks for fresh-n45-rand-02. It certified codimension at least **46,472** and fresh exhaustive enumeration found **0** solutions. Thus the strict inequality establishes **LFD at least 5** for this named Boolean presentation.

The certificate is 1,045,099,360 bytes with SHA-256 e3ec47c17f44e25dc541b4dd6abebe2fa342bb1d186d0499e3763e5c3d2ea616. The collector checked all 47 worker artifacts and successful terminal return; three controls passed. Public certificate and evidence readback were verified.



Admission passed after 50.114 seconds at 48,864,034,816 bytes available. Producer and checker took 1987.561 and 480.203 seconds. The original panel now has 30 terminal, 23 certified LFD-at-least-five, and two remaining inputs. Frozen source 608be655740605d96d97157006c29894332a72db; protocol 5936dd6d08dded341aac3de83a3f88766b6c2535185fcad5cf9ca720dbce4c73.

Input n=45, k=23, seed 12558114396312731448, degree 4. Runtime command and paths are in report.md and the frozen plan. CPU time was not recorded.